

PROJECT TITLE : AI FAQ ASSISTANT API

PROJECT DEVELOPMENT PHASE

PROJECT SETUP:

The project was set up using Visual Studio Code by creating the project folder and `src` directory. The required packages such as Express, Mongoose, dotenv, CORS, JSON Web Token, bcrypt, and Google GenAI were installed using npm.

BACKEND DEVELOPMENT:

The backend was developed using Node.js and Express.js. The Express server manages API requests, routing, JSON data, CORS, and environment configuration. The application is organized into separate models, controllers, routes, and middleware.

AUTHENTICATION:

JWT-based authentication was implemented to secure private APIs. Authentication middleware verifies user tokens, while bcrypt is used to securely hash passwords.

DATABASE DEVELOPMENT:

MongoDB was integrated using Mongoose ODM. User and FAQ schemas were created with validation rules, timestamps, and relationships. FAQ records are linked to their creators using the `createdBy` field.

FAQ AND API DEVELOPMENT:

RESTful APIs were developed for registration, login, FAQ CRUD operations, FAQ search, and AI generation. Protected FAQ operations require a valid JWT token.

AI INTEGRATION:

Google Gemini AI was integrated using the Google GenAI SDK to generate FAQ questions, answers, and topic suggestions based on the given topic.

ERROR HANDLING AND VALIDATION:

Centralized error handling was implemented to provide consistent API responses. Mongoose schema validation ensures that valid and structured data is stored in MongoDB.

